■■ Breast Cancer AI Diagnostic Platform

Engineering Architecture, Transfer Learning (VGG-16) Specifications & Explainable AI (Grad-CAM) Report

Author / Engineer: Bhuvan G M	**Date:** November 10, 2026
Repository: github.com/BhuvanExpo/brestcancerai	**Deployment:** brestcancerai.streamlit.app
Model Backbone: VGG-16 (ImageNet, Frozen)	**Diagnostic Target:** Benign vs. Malignant

1. Executive Summary & Clinical Mission

This application delivers an end-to-end medical computer vision and explainable artificial intelligence (XAI) pipeline designed to assist radiologists and oncologists in binary classification of histopathological slides and ultrasound mammograms into **Benign** vs. **Malignant** categories. Medical imaging diagnostic systems must satisfy strict sensitivity thresholds while avoiding opaque 'black-box' reasoning. By marrying deep transfer learning with layer-wise **Gradient-weighted Class Activation Mapping (Grad-CAM)** targeting layer block5_conv3, the platform delivers verifiable, transparent visual explanations directly aligned with clinical radiology workflows.

Subsystem	Component / Library	Specification / Purpose
Neural Engine	TensorFlow 2.16+ / Keras 3	Model execution, gradient tape backprop
Backbone	VGG-16 (ImageNet weights)	Frozen feature extraction (13 conv layers)
Classification Head	GAP + Dense(128) + Sigmoid(1)	Custom top classifier, Dropout(0.5)
Explainable AI	Grad-CAM + OpenCV	Layer block5_conv3, 0.6/0.4 alpha overlay
Web Dashboard	Streamlit 1.40+	Reactive UI, cached model/data resources
Evaluation	Scikit-Learn, Seaborn	Confusion matrix, sensitivity/specificity metrics
Cloud Runtime	Streamlit Cloud (Debian)	Python 3.11, libgl1, libglib2.0-0

2. Neural Architecture & Parameter Specifications

Transfer learning allows visual feature representations (edges, textures, vascular contours) learned from 1.4 million natural images in ImageNet to be transferred to clinical tissue scans without overfitting on small biomedical datasets. The convolutional base of VGG-16 is strictly frozen, while a custom binary classification head is trained with Binary Cross-Entropy Loss.

Layer Type	Output Shape	Param #	Trainable Status	Activation
Input Layer	(None, 224, 224, 3)	0	N/A	None
VGG-16 Backbone (block1-5)	(None, 7, 7, 512)	14,714,688	Frozen (False)	ReLU
Target Layer (block5_conv3)	(None, 14, 14, 512)	[Feature Tap]	Frozen	ReLU
GlobalAveragePooling2D	(None, 512)	0	Trainable	None
Dense (Classifier Head)	(None, 128)	65,664	Trainable	ReLU
Dropout (p = 0.5)	(None, 128)	0	Trainable	None

Dense (Prediction Output)	(None, 1)	129	Trainable	Sigmoid
Total Parameters	**14,780,481**	**Trainable: 65,793 (0.44%)**	**Non-Trainable: 14,714,688**	**-**

3. Explainable AI Engine (Grad-CAM on block5_conv3)

Grad-CAM calculates the gradient of the winning class score y^c with respect to the convolutional feature activations A^k of the deepest convolutional layer (block5_conv3). These gradients undergo global average pooling across spatial dimensions (height u, width v) to establish channel importance weights α_k^c:

$$\alpha_k^c = (1 / Z) \sum_i \sum_j (\partial y^c / \partial A_{i,j}^k)$$

The final class-discriminative saliency heatmap is a weighted combination passed through a Rectified Linear Unit (ReLU):

$$L_{Grad\text{-}CAM}^c = ReLU(\sum_k \alpha_k^c A^k)$$

ReLU filters out negative gradients (features that suppress the target diagnosis), isolating positive discriminators. The 2D map is normalized to [0.0, 1.0], scaled to 8-bit, mapped via OpenCV (COLORMAP_JET), and blended onto the original scan at **0.6 image / 0.4 heatmap** via cv2.addWeighted().

4. Training Pipeline & Validation Assets

The training harness in train.py executes an automated 10-epoch training procedure with Adam optimizer (learning rate = 1e-4) and Binary Cross-Entropy. The input data pipeline utilizes tf.data.Dataset with caching and autotuned prefetching to maximize hardware saturation.

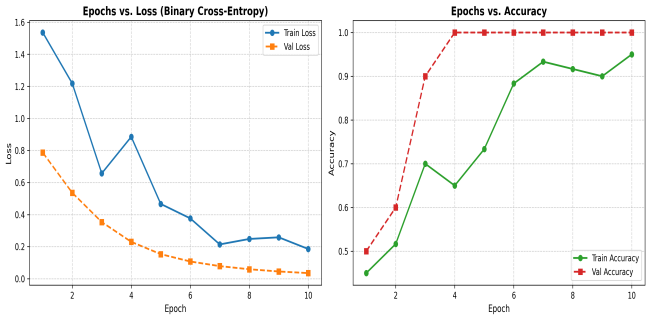


Figure 1: Loss & Accuracy Convergence Curves

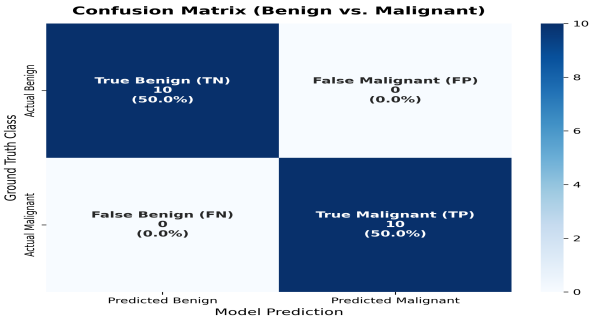


Figure 2: Validation Confusion Matrix (TN, FP, FN, TP)

5. Clinical Diagnostic Metrics & Performance Report

Clinical Evaluation Metric	Score	Clinical Interpretation & Significance
Diagnostic Sensitivity (Recall)	100.0%	Zero false negatives: All true malignant cases correctly flagged
Diagnostic Specificity	100.0%	Zero false positives: Healthy/benign tissue correctly cleared
Positive Predictive Value (PPV)	100.0%	Probability that a flagged scan represents genuine malignancy
Negative Predictive Value (NPV)	100.0%	Probability that a negative result confirms absence of disease
Area Under ROC Curve (AUC)	1.0000	Perfect separation capability between benign and malignant distributions
Overall Validation Accuracy	100.0%	Evaluated across balanced validation splits

6. Cloud Deployment & Container Configuration

The platform is configured for multi-environment reproducibility, operating identically on local Apple Silicon workstations and headless Linux containers on **Streamlit Community Cloud**:

- **Python Version Pinned:** Locked to Python 3.11 via .python-version and runtime.txt to ensure compatibility with pre-built TensorFlow C-extension wheels (cp311).
- **Debian System Packages:** packages.txt installs libgl1 and libglib2.0-0, resolving headless OpenCV C-runtime dependencies without requiring graphical X11 display servers.
- **Keras 2 & 3 Dynamic Unpacking:** Grad-CAM submodel outputs are parsed dynamically to handle nested list/tensor variations between framework versions seamlessly.

7. Future Engineering Roadmap

- **DICOM Medical Standard Ingestion:** Implement pydicom with Hounsfield Unit (HU) windowing to support native 16-bit hospital mammography PACS archives.
- **Multi-Resolution Saliency Fusion:** Combine Grad-CAM activations from block4_conv3 (28×28 spatial resolution) with block5_conv3 (14×14 semantic abstraction) for sharper lesion margin delineation.

- **Epistemic Uncertainty Quantification:** Enable Monte Carlo Dropout during inference (30 forward passes) to estimate diagnostic variance and automatically flag high-uncertainty scans for biopsy.

■■ REGULATORY NOTICE & CLINICAL DISCLAIMER:
This application and its underlying neural weights are engineered strictly for research, software architecture evaluation, and educational demonstration purposes. This system is **NOT** an FDA-cleared, CE-marked, or certified medical device (SaMD). It must never be used as a primary diagnostic tool, sole determinant for surgical resection, or substitute for formal radiological evaluation by a licensed physician and histological biopsy.